Operating Systems
(Honor Track)

File System 2: File System Case Studies, Buffering

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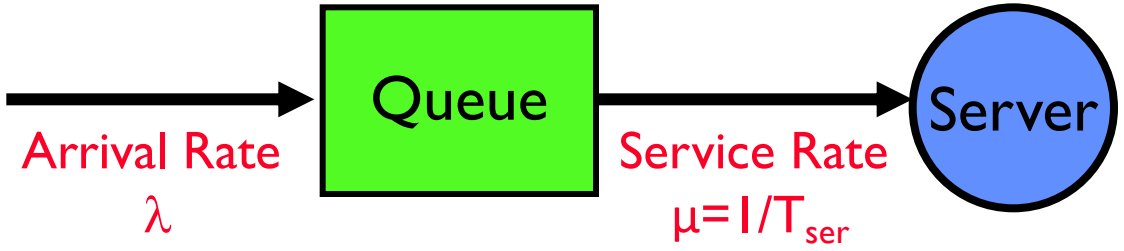
Acknowledgments: Ion Stoica, Berkeley CS 162

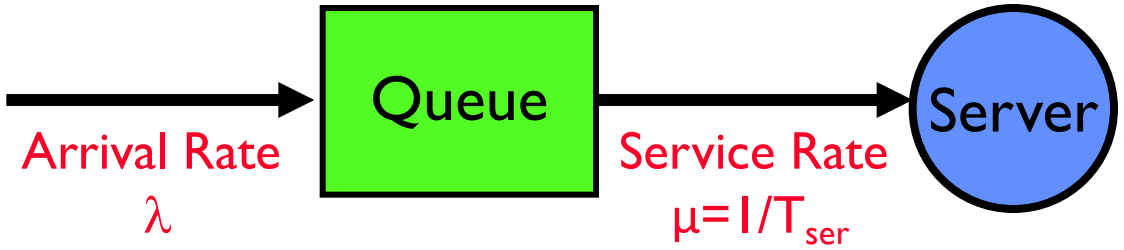
Recap: A Little Queuing Theory: Some Results (1/2)

- Assumptions:
  - System in equilibrium; No limit to the queue
  - Time between successive arrivals is random and memoryless



- Parameters that describe our system:
  - λ: mean number of arriving customers/second
  - T_{ser}: mean time to service a customer ("m")
  - C: squared coefficient of variance = σ^2/m^2
  - μ: service rate = $1/T_{ser}$
  - u: server utilization ($0 \le u \le 1$): $u = \lambda/\mu = \lambda \times T_{ser}$
- Parameters we wish to compute:
  - T_q: Time spent in the queue
  - L_q: Length of queue = $\lambda \times T_q$ (by Little's law)

Recap: A Little Queuing Theory: Some Results (2/2)



- Parameters that describe our system:
  - λ: mean number of arriving customers/second $\lambda = 1/T_A$
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- Parameters we wish to compute:
  - T_q: Time spent in the queue
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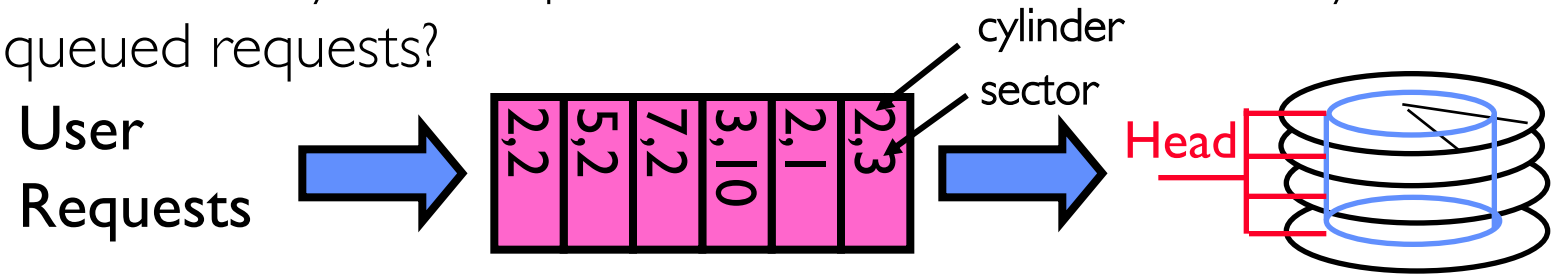
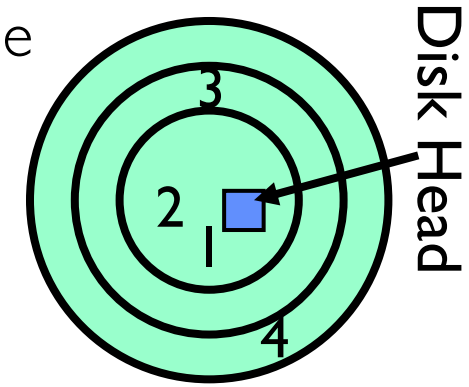
- **Results** (**M**: Poisson arrival process, **1** server):
  - **M**emoryless service time distribution (C = 1): Called an **M/M/1** queue
    - » $T_q = T_{ser} \times \boxed{u/(1-u)}$
  - **G**eneral service time distribution (no restrictions): Called an **M/G/1** queue
    - » $T_q = T_{ser} \times \frac{1}{2}(1+C) \times \boxed{u/(1-u)}$

Recap: Disk Scheduling (1/3)

- Disk can do only one request at a time; What order do you choose to do queued requests?

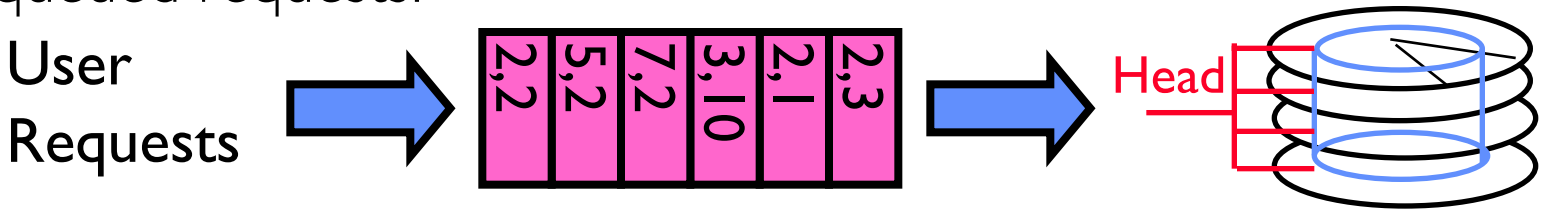


- FIFO Order
  - Fair among requesters, but order of arrival may be to random spots on the disk ⇒ Very long seeks
- SSTF: Shortest seek time first
  - Pick the request that's closest on the disk
  - Although called SSTF, today must include rotational delay in calculation, since rotation can be as long as seek
  - Con: SSTF good at reducing seeks, but may lead to starvation

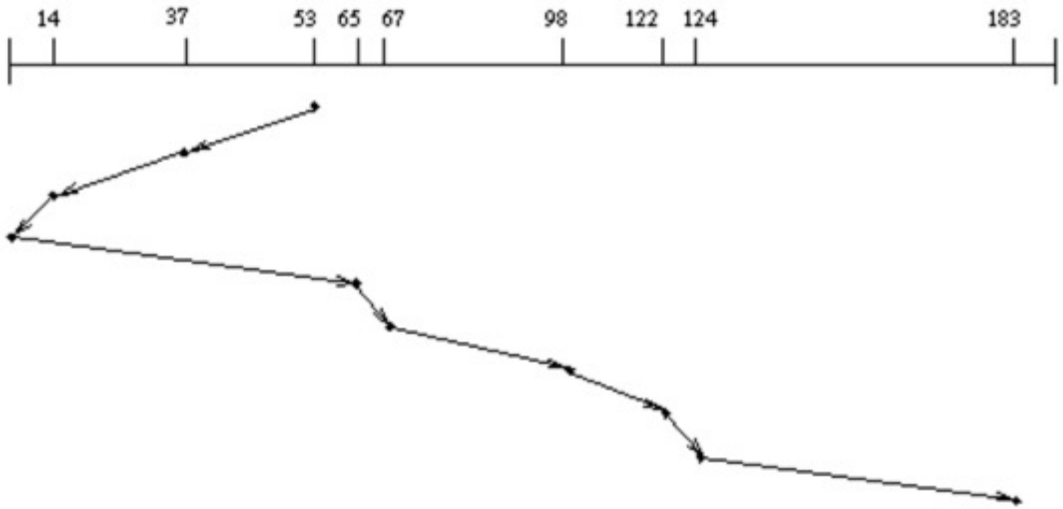


Recap: Disk Scheduling (2/3)

- Disk can do only one request at a time; What order do you choose to do queued requests?



- SCAN: Implements an Elevator Algorithm: take the closest request in the direction of travel
  – No starvation, but retains flavor of SSTF



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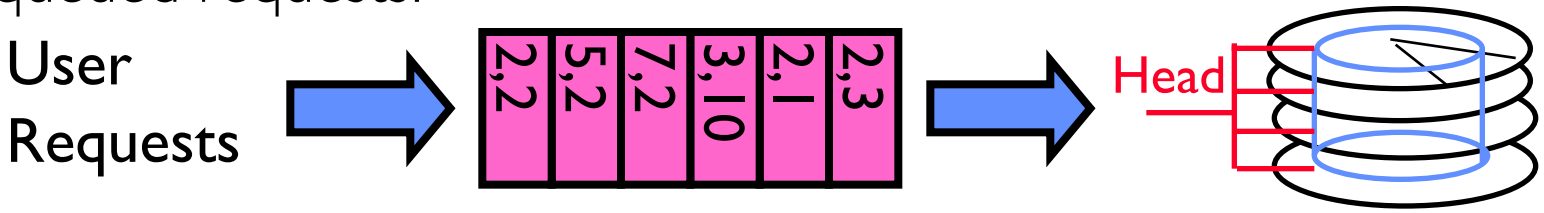
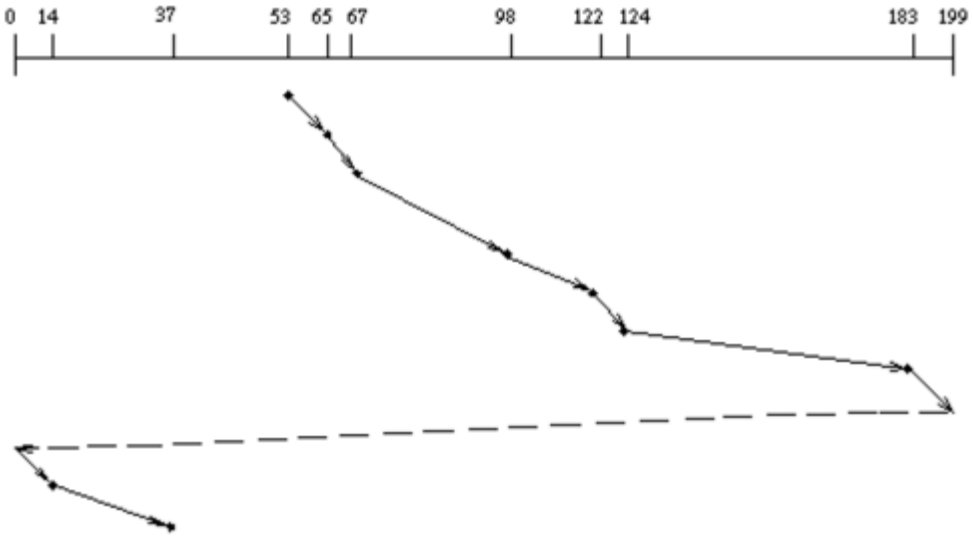
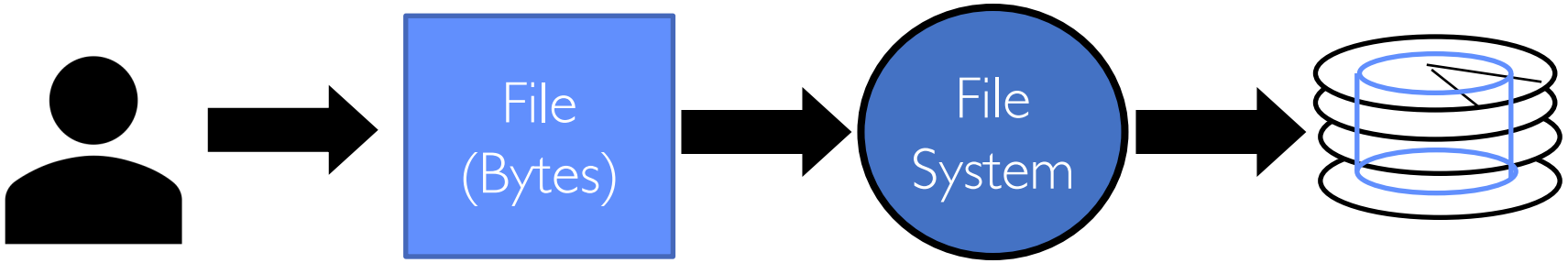


- C-SCAN: Circular-Scan: only goes in one direction
  - Skips any requests on the way back
  - Fairer than SCAN, not biased towards pages in middle



Recap: Translation from User to System View



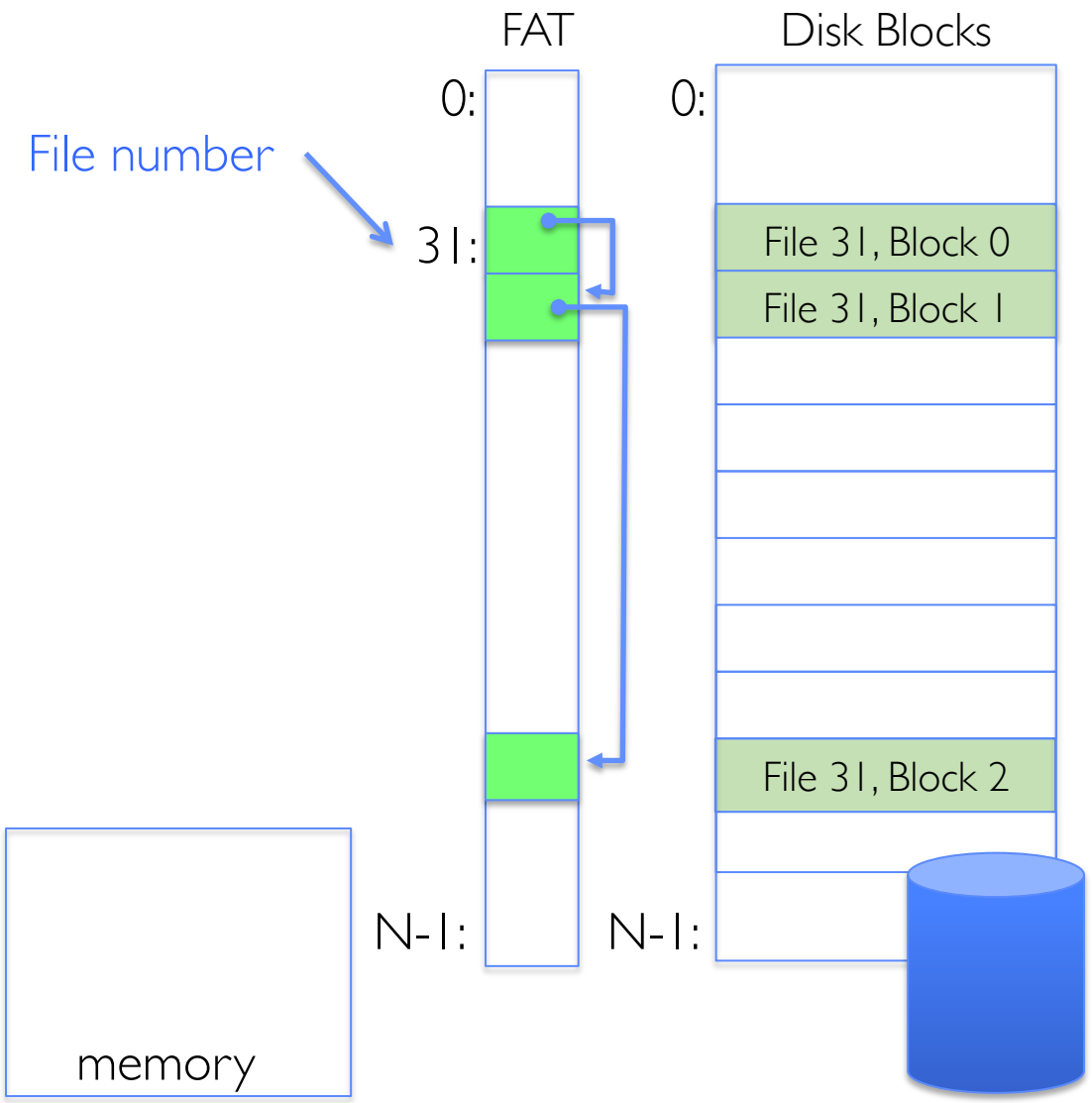
- What happens if user says: "give me bytes 2 – 12?"
  - Fetch block corresponding to those bytes
  - Return just the correct portion of the block
- What about writing bytes 2 – 12?
  - Fetch block, modify relevant portion, write out block
- Everything inside file system is in terms of whole-size blocks
  - Actual disk I/O happens in blocks
  - read/write smaller than block size needs to translate and buffer

CASE STUDY:
FAT: FILE ALLOCATION TABLE

- MS-DOS, 1977
- Still widely used!

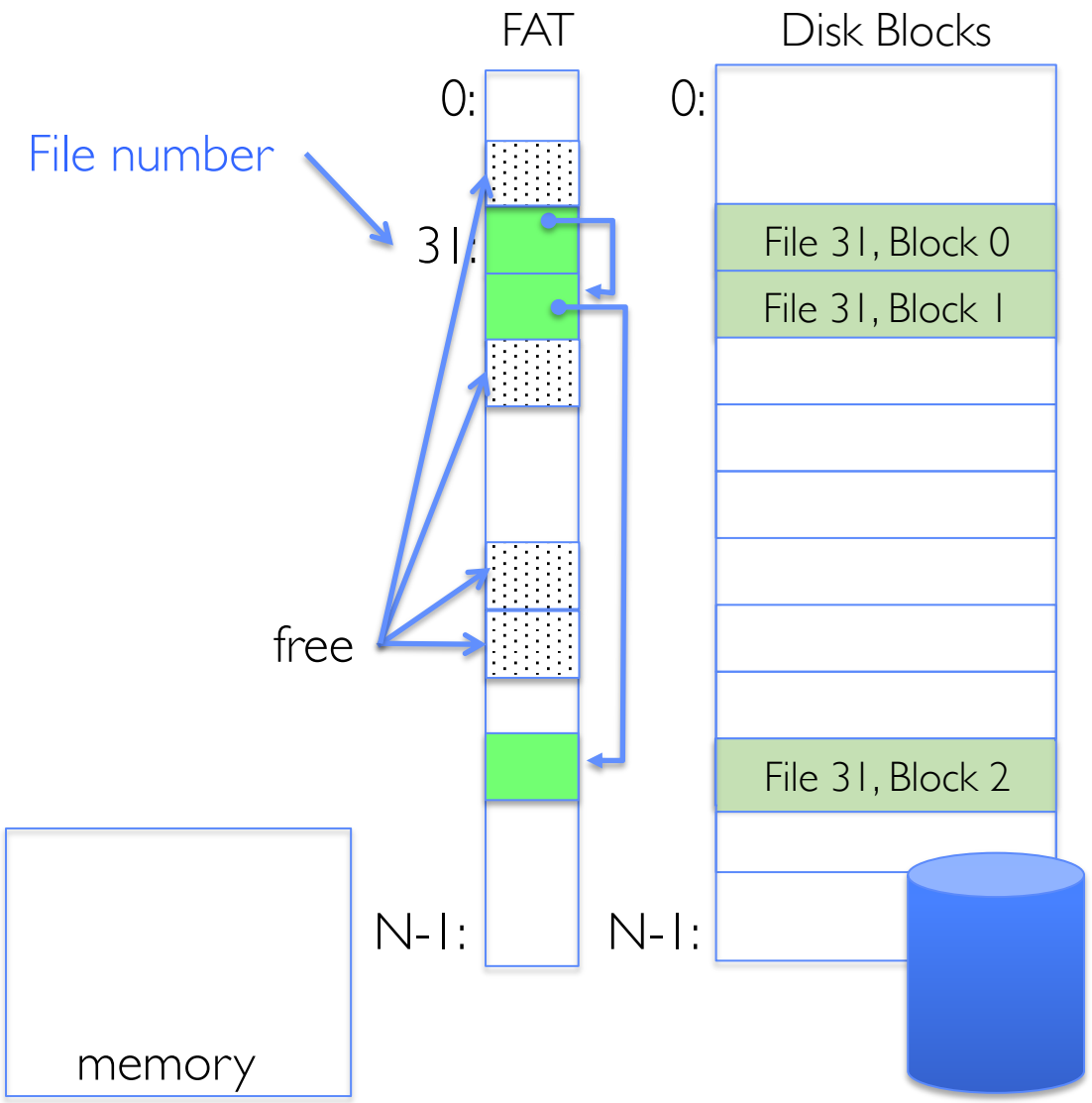
FAT (File Allocation Table)

- Assume (for now) we have a way to translate a path to a "file number"
  - i.e., a directory structure
- Disk Storage is a collection of Blocks
  - Just hold file data (offset o = < B, x >)
- Example: file_read 31, < 2, x >
  - Index into FAT with file number
  - Follow linked list to block
  - Read the block from disk into memory



FAT (File Allocation Table)

- File is a collection of disk blocks
- FAT is a linked list with blocks
- File number is index of root of block list for the file
- File offset: block number and offset within block
- Follow list to get block number
- Unused blocks marked free
  - Could require scan to find
  - Or, could use a free list



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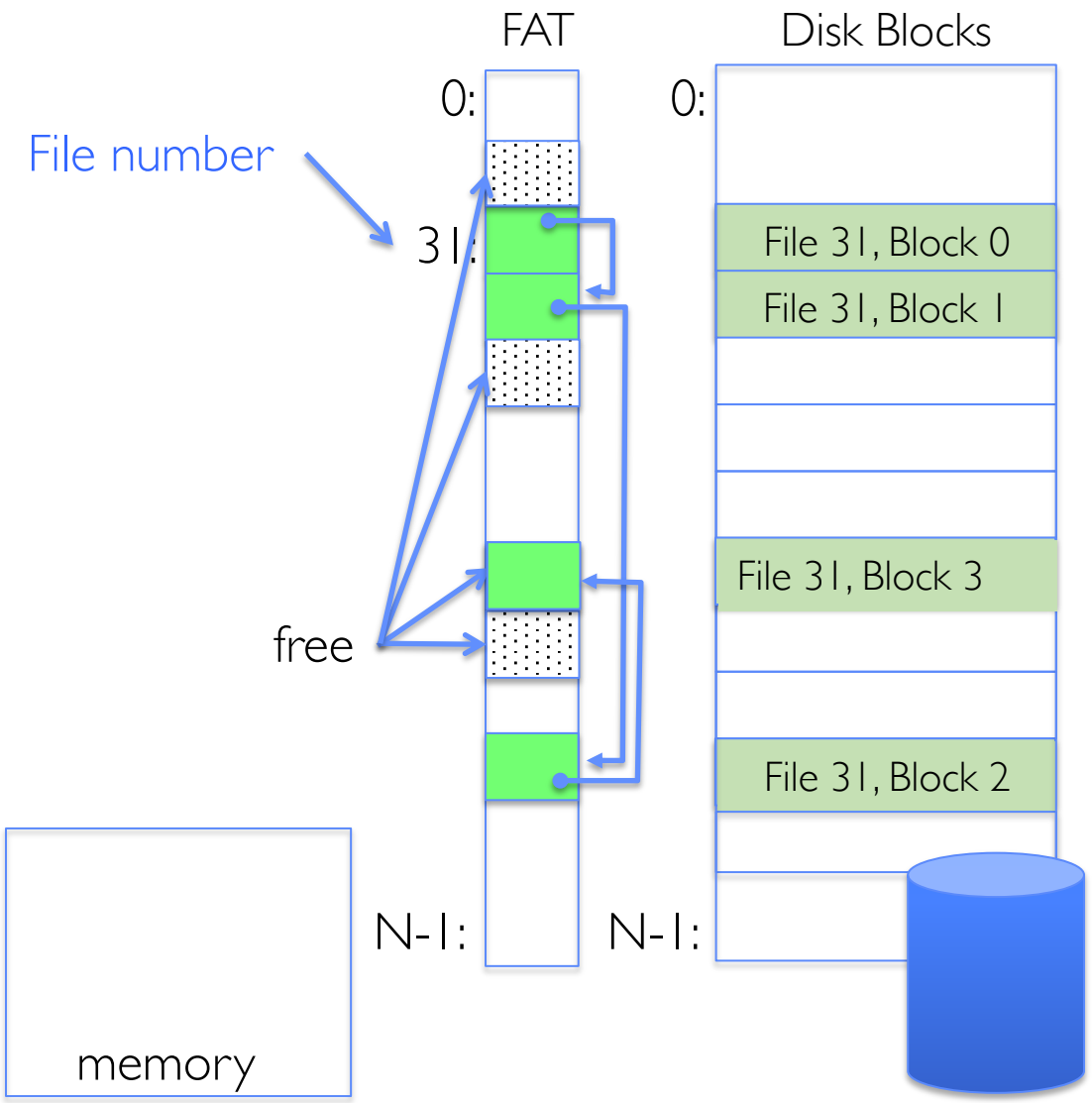
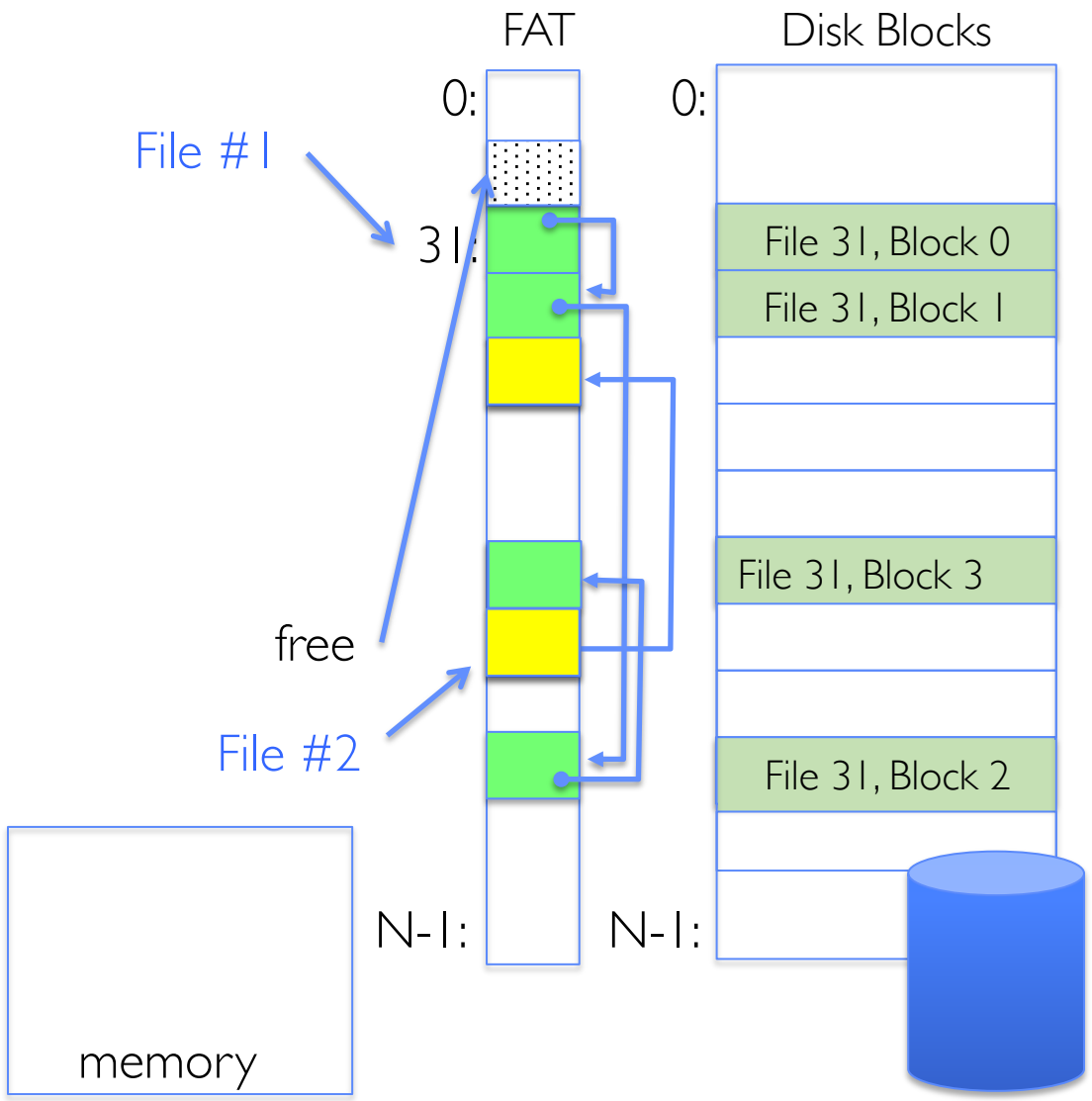
- Ex: file_write(31, < 3, y >)
  - Grab free block
  - Linking them into file



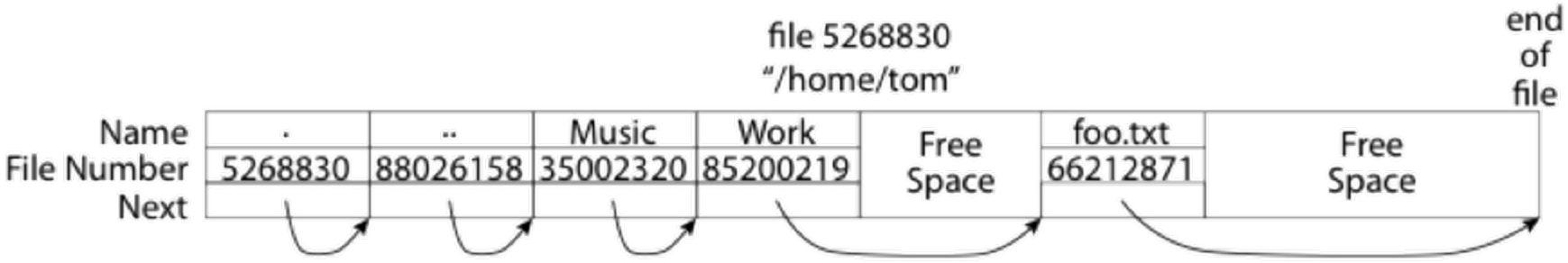
FAT (File Allocation Table)

- Where is FAT stored?
  - On disk
- How to format a disk?
  - Zero the blocks, mark FAT entries "free"
- How to quick format a disk?
  - Mark FAT entries "free"

- Simple: can implement in device firmware



FAT: Directories

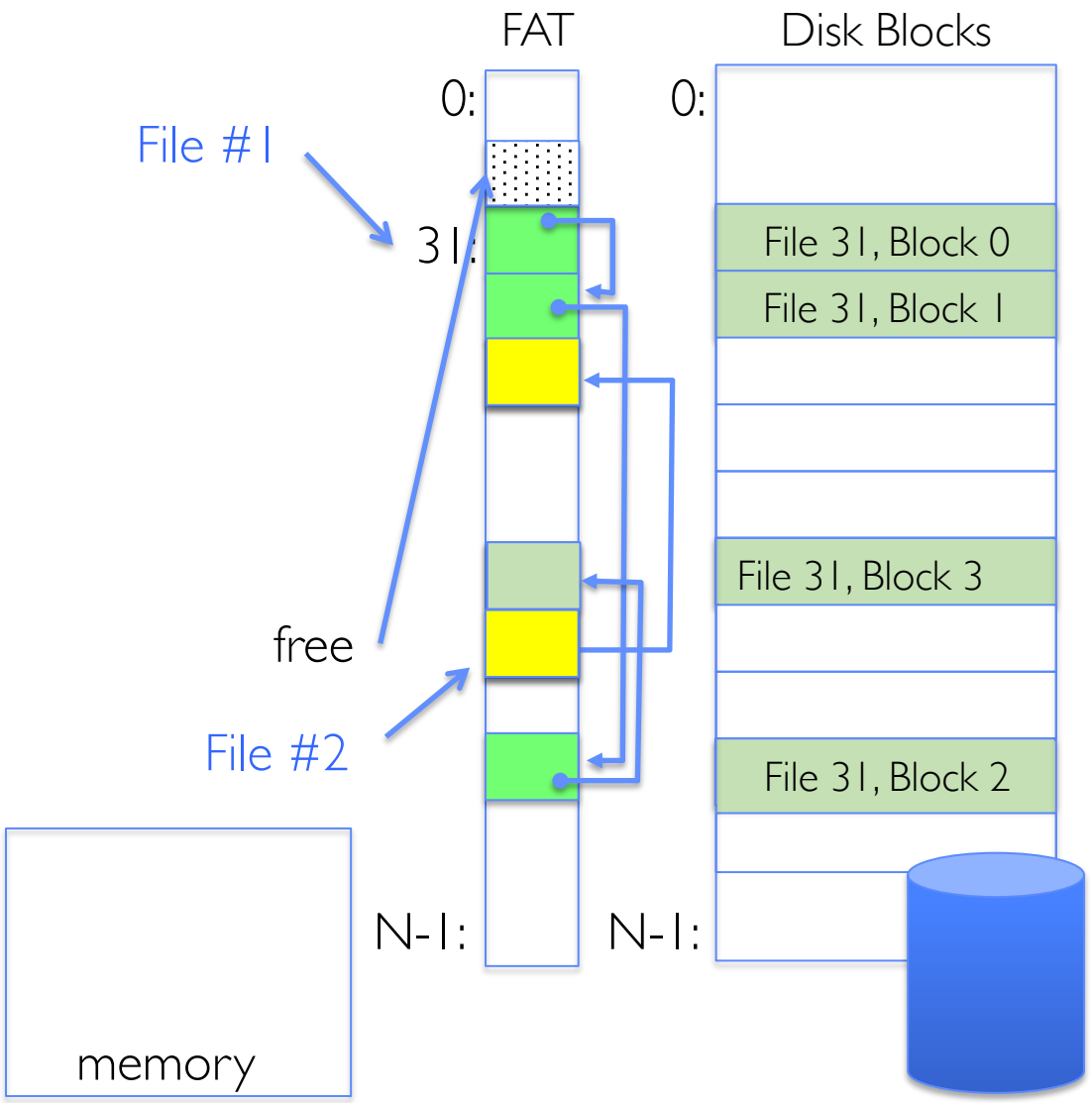


- A directory is a file containing <file_name: file_number> mappings
- Free space for new/deleted entries
- In FAT: file attributes are kept in directory (!!!)
  - Not directly associated with the file itself
- Each directory is a linked list of entries
  - Requires linear search of directory to find particular entry
- Where do you find root directory ("/")?
  - At well-defined place on disk
  - For FAT, this is at block 2 (there are no blocks 0 or 1)

FAT Discussion

Suppose you start with the file number:

- Time to find block?
- Block layout for file?
- Sequential access?
- Random access?
- Fragmentation?
- Small files?
- Big files?



CASE STUDY:
UNIX FILE SYSTEM (BERKELEY FFS)

Inodes in Unix (Including Berkeley FFS)

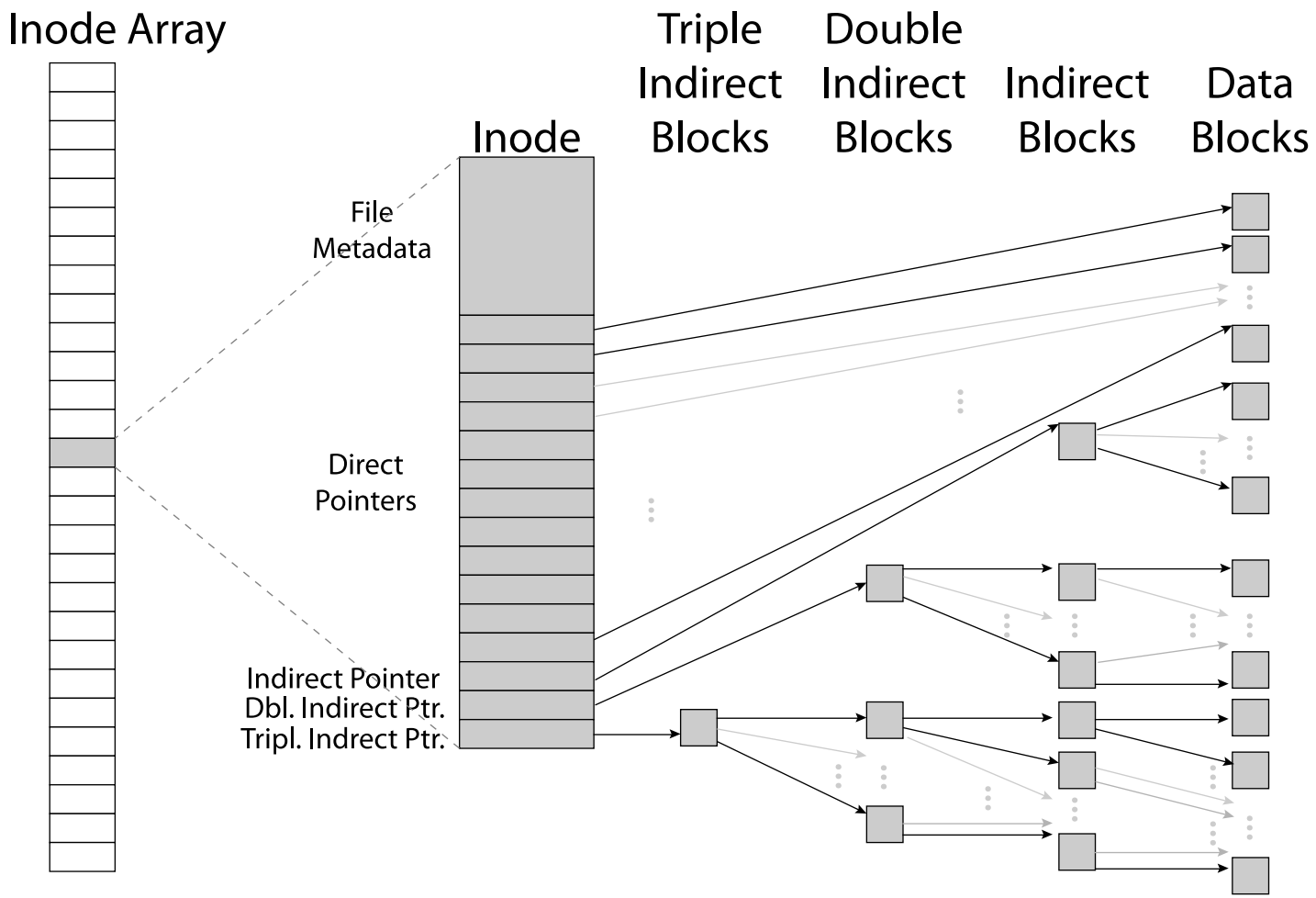
- File Number (inumber) is index into an array of inodes (index structure)

- Each inode corresponds to a file and contains its metadata
  - So, things like read/write permissions are stored with *file,* not in directory (like in FAT)
  - Allows multiple names (directory entries) for a file

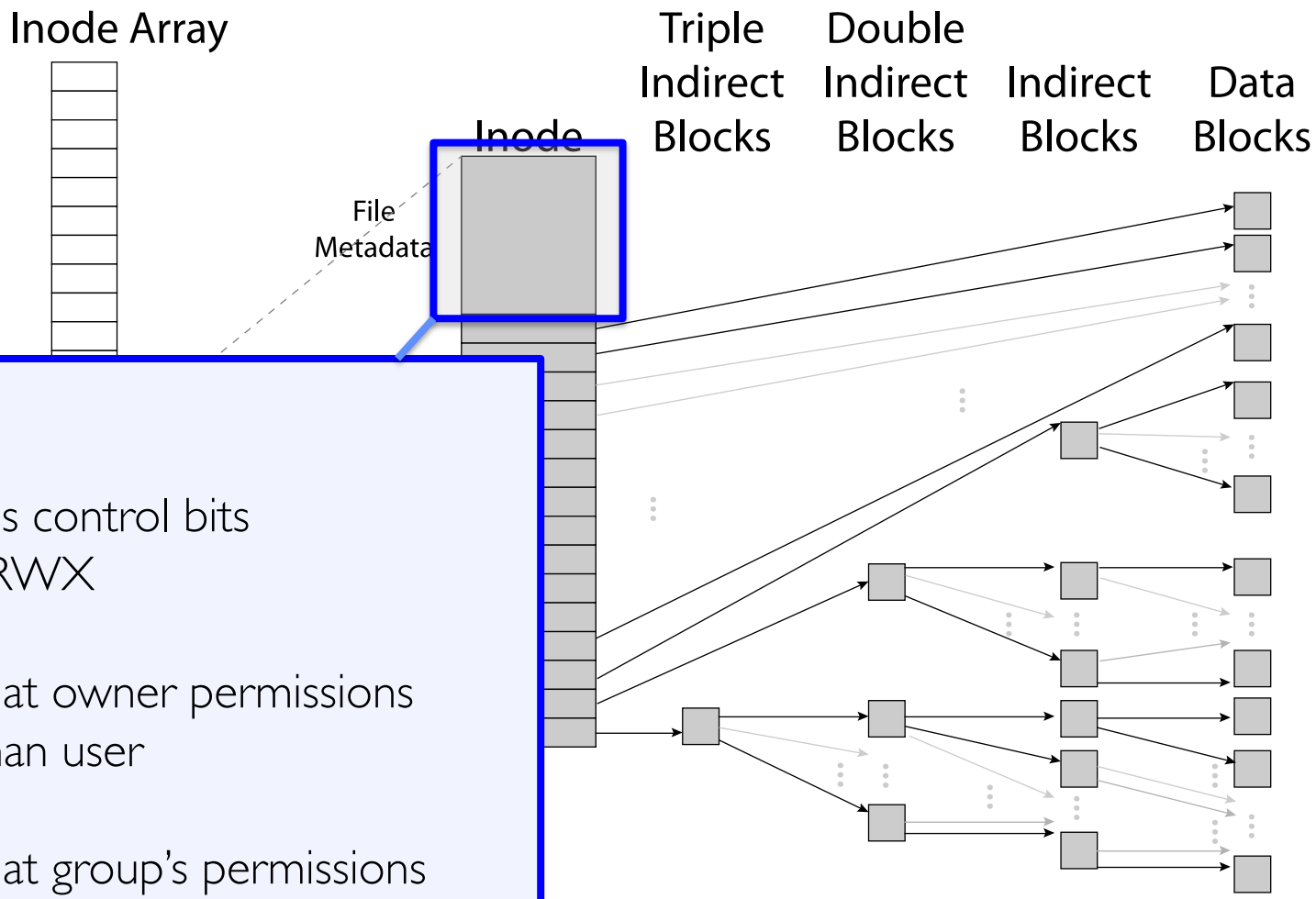
- Inode maintains a multi-level tree structure to find storage blocks for files
  - Great for little and large files
  - Asymmetric tree with fixed sized blocks

- Original ***inode*** format appeared in BSD 4.1 (more following)
  - Berkeley Standard Distribution Unix!
  - Similar structure for Linux Ext 2/3

Inode Structure



File Atributes

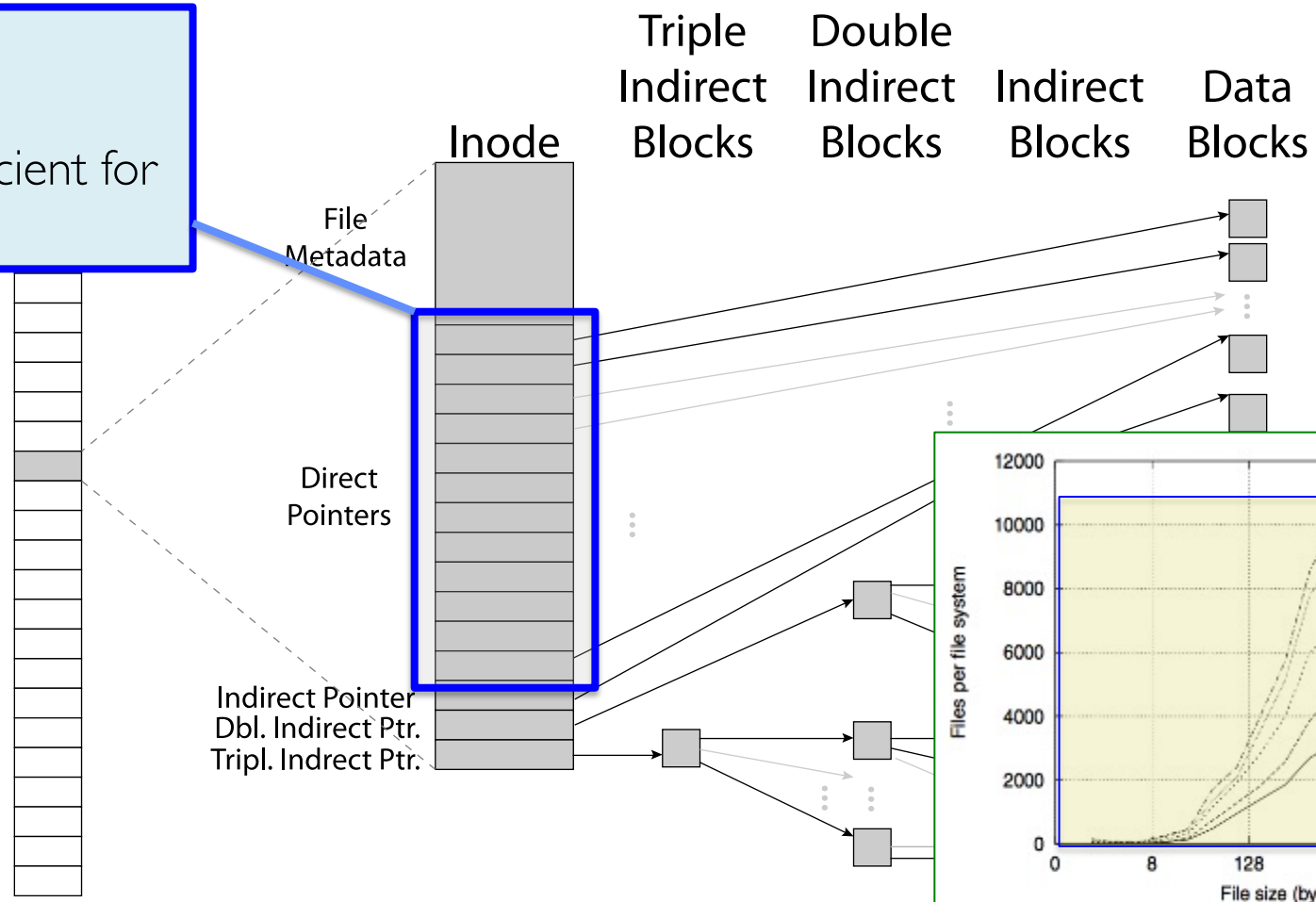


User
Group
9 basic access control bits
- UGO x RWX
SetUID bit
- execute at owner permissions rather than user
SetGID bit
- execute at group's permissions

Small Files: 12 Pointers Direct to Data Blocks

Direct pointers

4KB blocks ⇒ sufficient for files up to 48KB



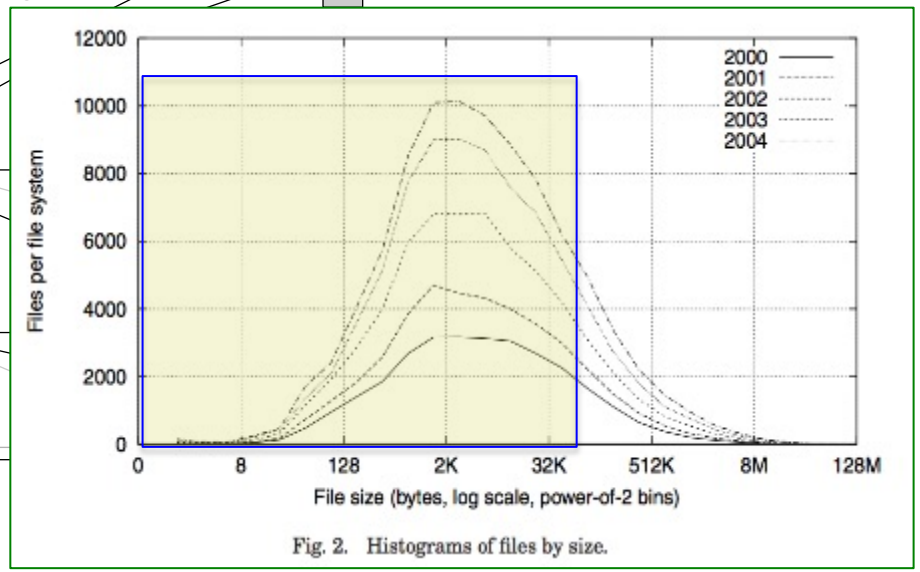


Fig. 2. Histograms of files by size.

Large Files: 1-, 2-, 3-level indirect pointers

Indirect pointers
- point to a disk block containing only pointers
- 4 KB blocks => 1024 ptrs
  => 4 MB @ level 2
  => 4 GB @ level 3
  => 4 TB @ level 4

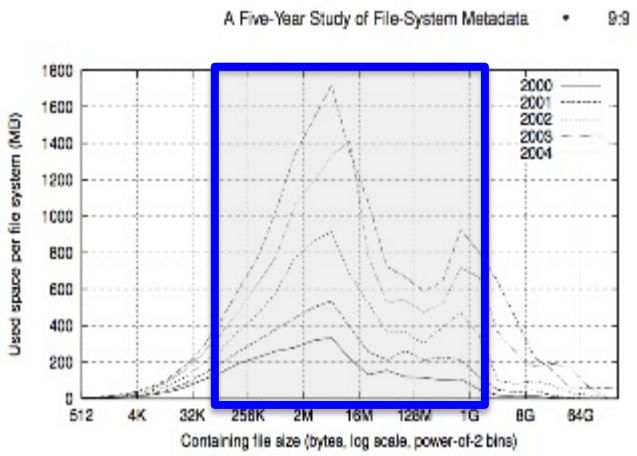
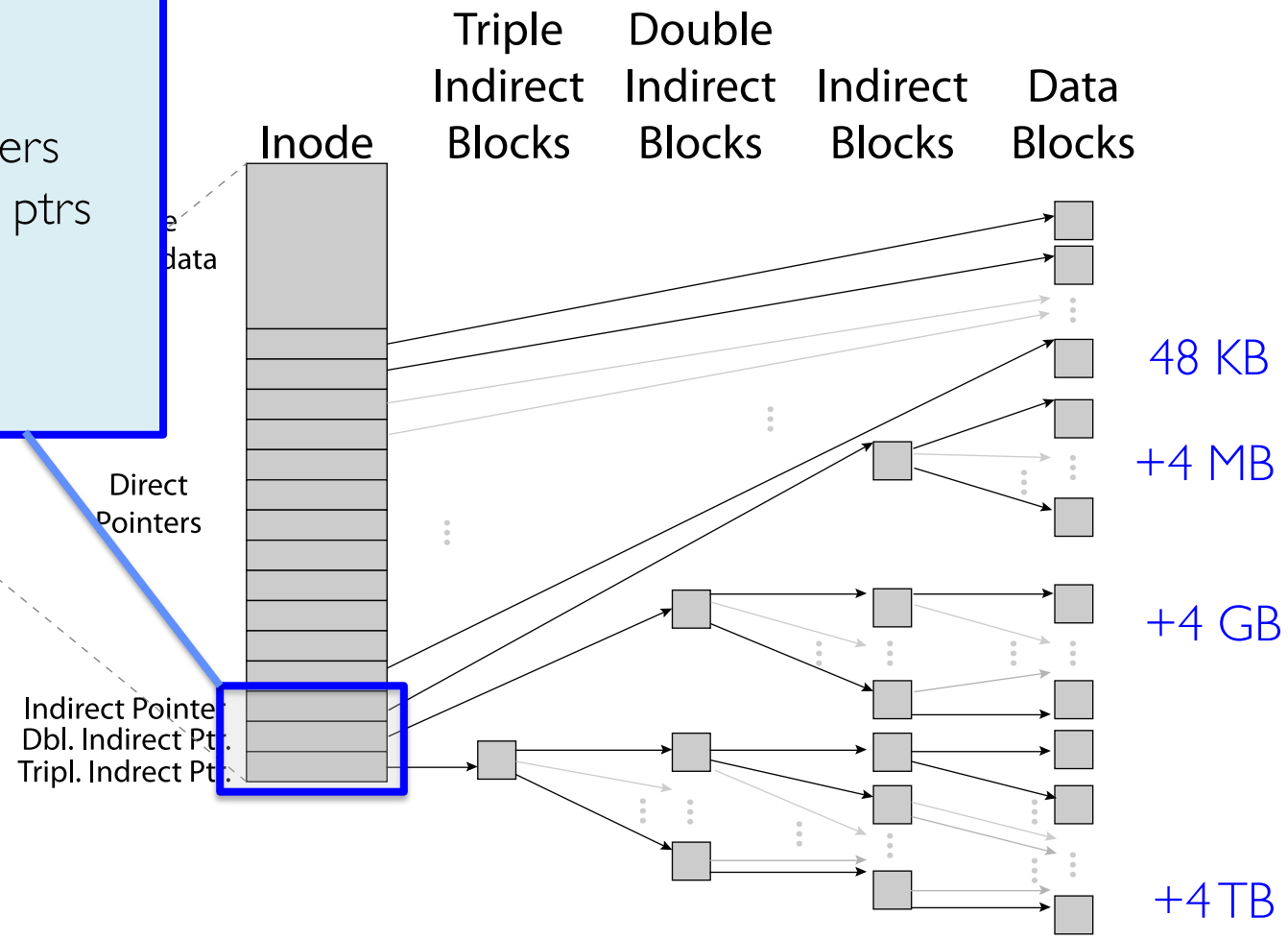
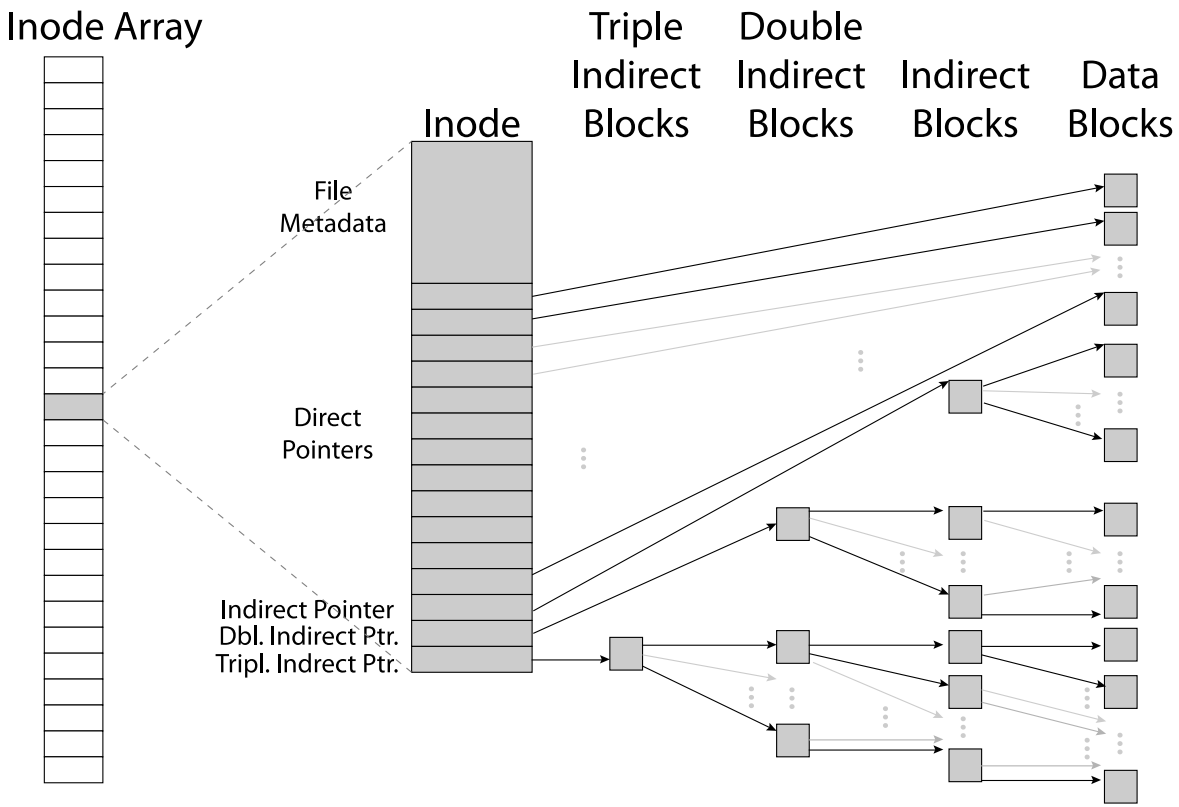


Fig. 4. Histograms of bytes by containing file size.



Putting it All Together: On-Disk Index

- Sample file in multilevel indexed format:
  - 10 direct ptrs, 1KB blocks
    - » 256 indirect blocks
    - » 256^2 double indirect blocks
    - » 256^3 triple indirect blocks
  - How many accesses for block #23? (assume file header accessed on open)?
    - » Two: One for indirect block, one for data
  - How about block #5?
    - » One: One for data
  - Block #340?
    - » Three: double indirect block, indirect block, and data



CASE STUDY:
BERKELEY FAST FILE SYSTEM (FFS)

Fast File System (BSD 4.2, 1984)

- Same inode structure as in BSD 4.1
  - Same file header and triply indirect blocks like we just studied
  - Some changes to block sizes from 1024 $\Rightarrow$ 4096 bytes for performance
- Paper on FFS: "A Fast File System for UNIX"
  - Marshall McKusick, William Joy, Samuel Leffler and Robert Fabry

- Optimization for Performance and Reliability:
  - Distribute inodes among different tracks to be closer to data
  - Use bitmap allocation in place of freelist
  - Attempt to allocate files contiguously
  - 10% reserved disk space
  - Skip-sector positioning

FFS Changes in Inode Placement: Motivation

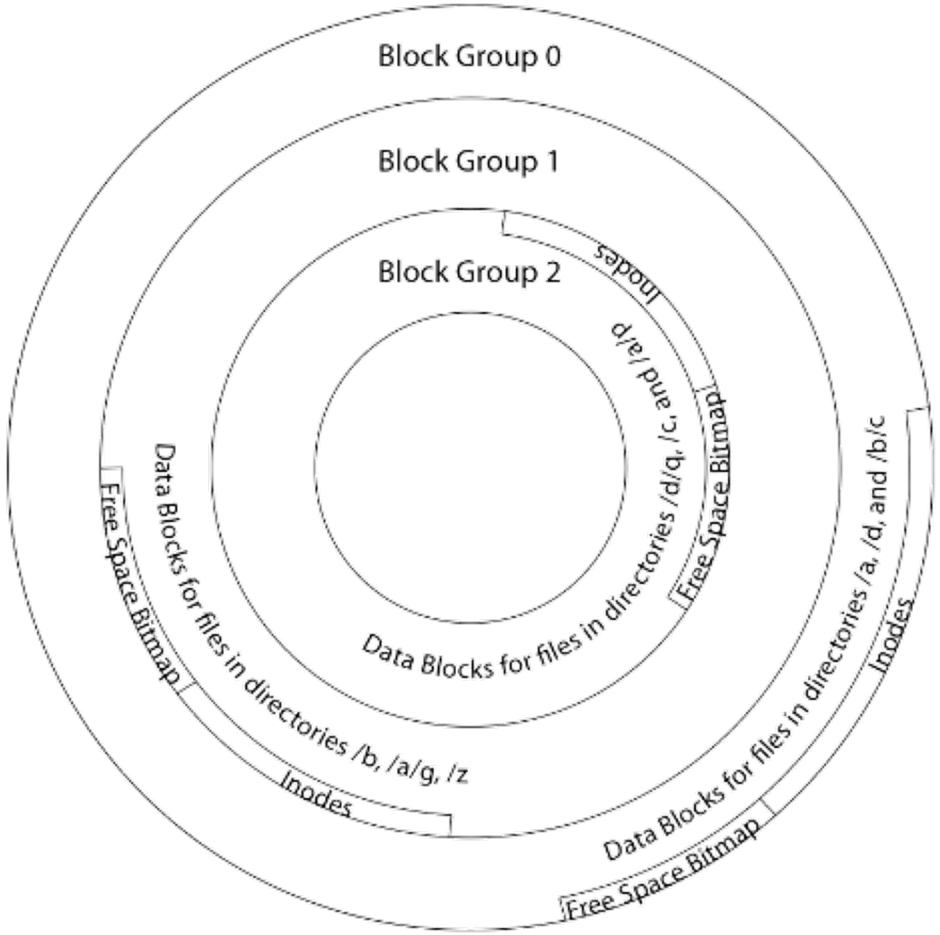
- In early UNIX and DOS/Windows FAT file system, headers stored in special array in outermost cylinders
  - Fixed size, set when disk is formatted
    - » At formatting time, a fixed number of inodes are created
    - » Each is given a unique number, called an "inumber"

- Problem #1: Inodes all in one place (outer tracks)
  - Head crash potentially destroys all files by destroying inodes
  - Inodes not close to the data that they point to
    - » To read a small file, seek to get header, seek to get data

- Problem #2: When create a file, don't know how big it will become (in UNIX, most writes are by appending)
  - How much contiguous space do you allocate for a file?
  - Makes it hard to optimize for performance

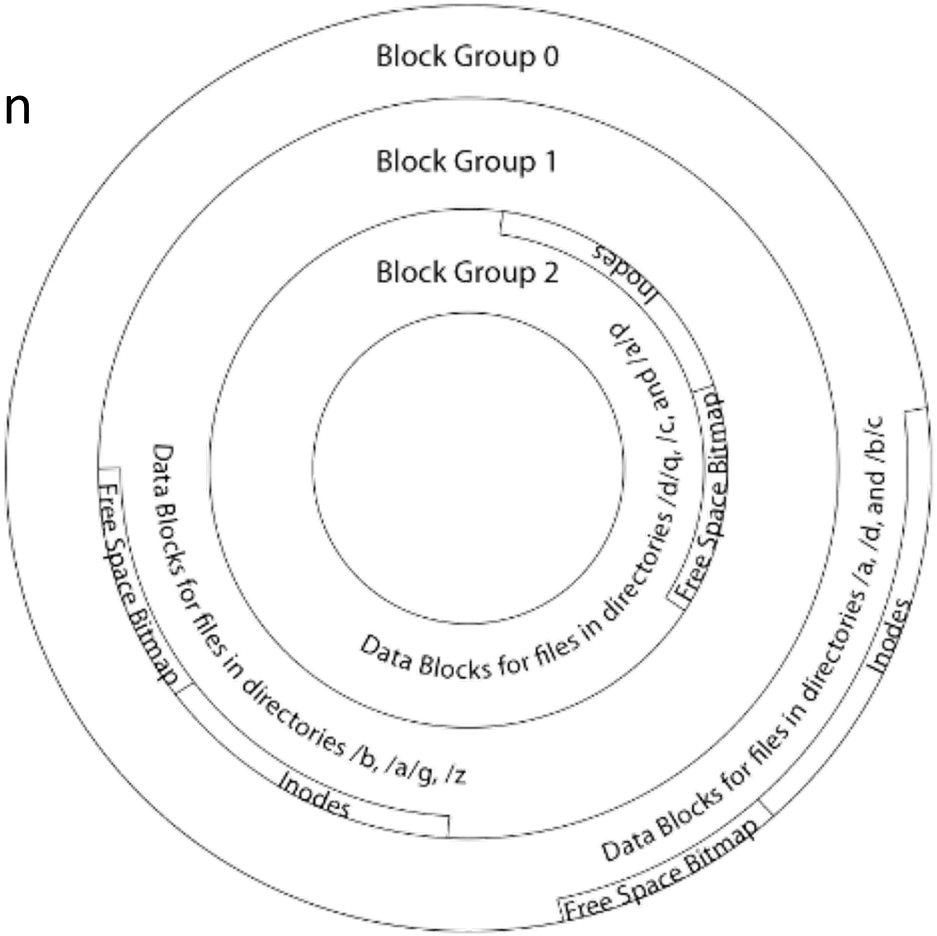
FFS Locality: Block Groups

- The UNIX BSD 4.2 (FFS) distributed the header information (inodes) closer to the data blocks
  - Often, inode for file stored in same ‘‘cylinder group’’ as parent directory of the file
  - makes an “ls” of that directory run very fast
- File system volume divided into set of block groups
  - Close set of tracks
- Data blocks, metadata, and free space interleaved within block group
  - Avoid huge seeks between user data and system structure
- Put directory and its files in common block group

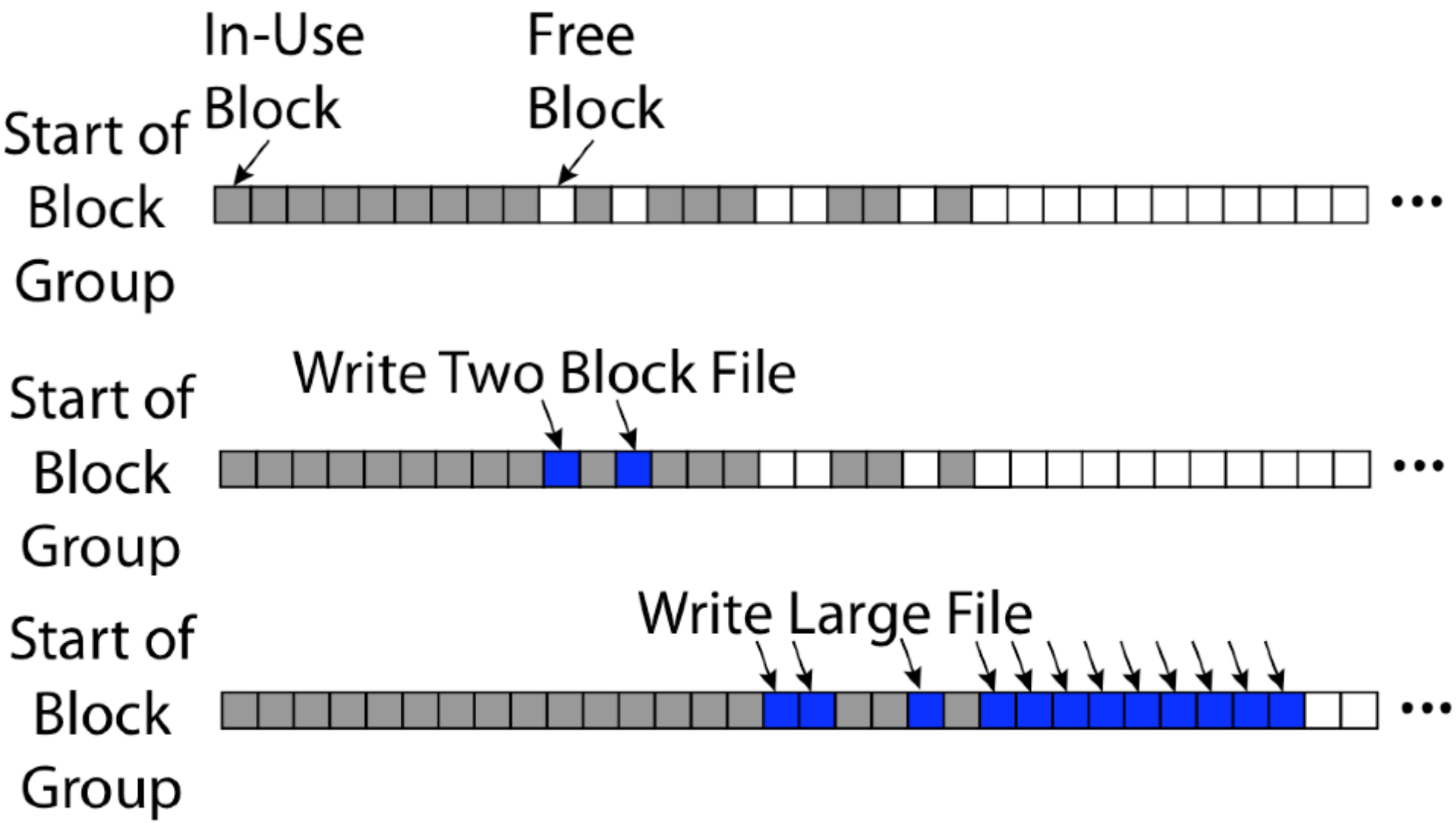


FFS Locality: Block Groups (Con't)

- First-Free allocation of new file blocks
  - To expand file, first try successive blocks in bitmap, then choose new range of blocks
  - Few little holes at start, big sequential runs at end of group
  - Avoids fragmentation
  - Sequential layout for big files
- Important: keep 10% or more free!
  - Reserve space in the Block Group
- Summary: FFS Inode Layout Pros
  - For small directories, can fit all data, file headers, etc. in same cylinder ⇒ no seeks!
  - File headers much smaller than whole block (a few hundred bytes), so multiple headers fetched from disk at same time
  - Reliability: whatever happens to the disk, you can find many of the files (even if directories disconnected)

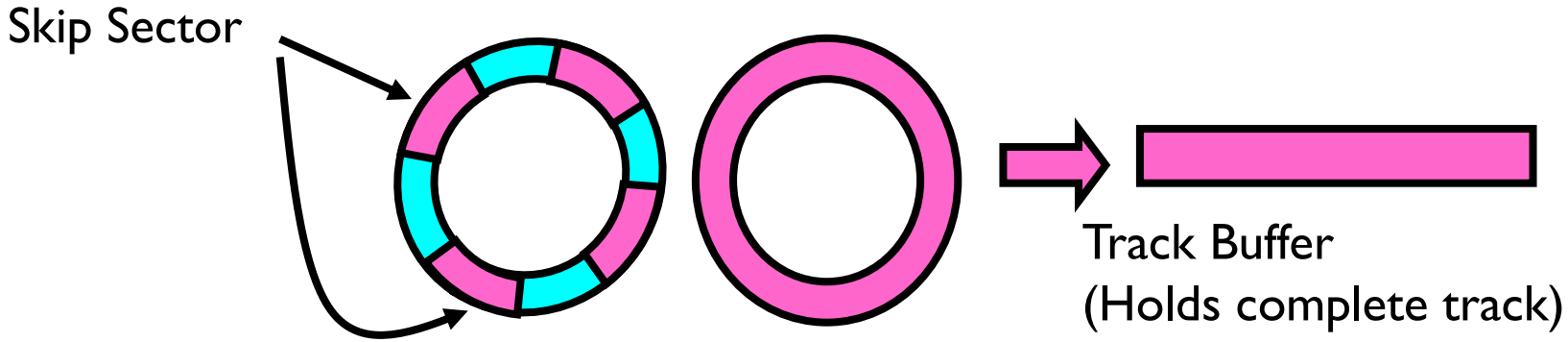


UNIX 4.2 BSD FFS First Fit Block Allocation



Attack of the Rotational Delay

- Problem 3: Missing blocks due to rotational delay
  - Issue: Read one block, do processing, and read next block. In meantime, disk has continued turning: missed next block! Need 1 revolution/block!



  - Solution1: Skip sector positioning ("interleaving")
    - » Place the blocks from one file on every other block of a track: give time for processing to overlap rotation
    - » Can be done by OS or in modern drives by the disk controller
  - Solution 2: Read ahead: read next block right after first, even if application hasn't asked for it yet
    - » This can be done either by OS (read ahead)
    - » By disk itself (track buffers) - many disk controllers have internal RAM that allows them to read a complete track
- Modern disks + controllers do many things "under the covers"
  - Track buffers, elevator algorithms, bad block filtering

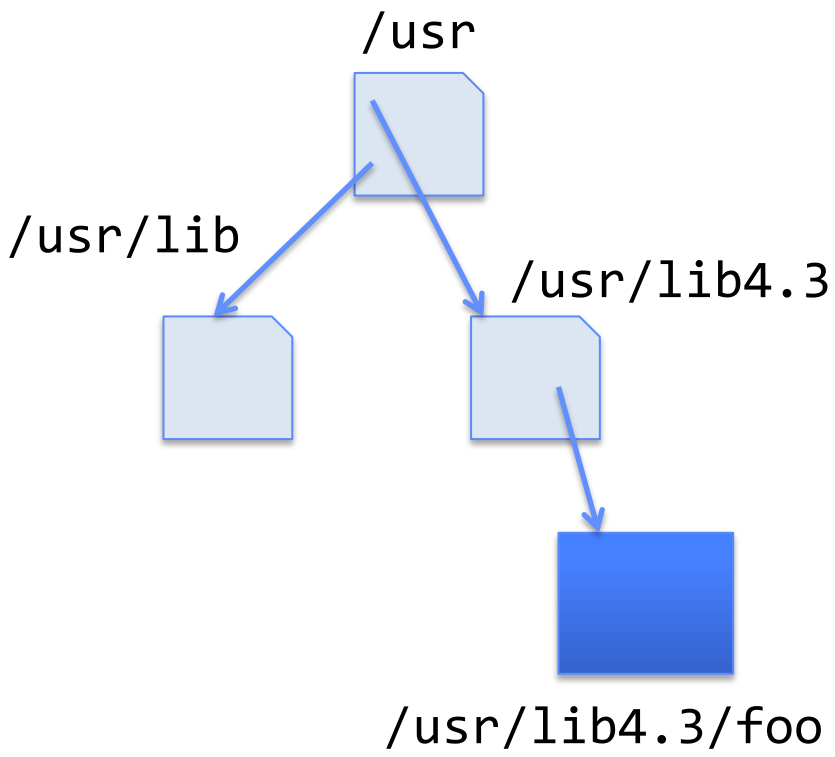
UNIX 4.2 BSD FFS

- Pros
  - Efficient storage for both small and large files
  - Locality for both small and large files
  - Locality for metadata and data
  - No defragmentation necessary!

- Cons
  - Inefficient for tiny files (a 1 byte file requires both an inode and a data block)
  - Inefficient encoding when file is mostly contiguous on disk
  - Need to reserve 10-20% of free space to prevent fragmentation

Hard Links

- Hard link
  - Mapping from name to file number in the directory structure
  - First hard link to a file is made when file created
  - Create extra hard links to a file with the link() system call
  - Remove links with unlink() system call
- When can file contents be deleted?
  - When there are no more hard links to the file
  - Inode maintains reference count for this purpose



Soft Links (Symbolic Links)

- Soft link or Symbolic Link or Shortcut
  - Directory entry contains the path and name of the file
  - Map one name to another name

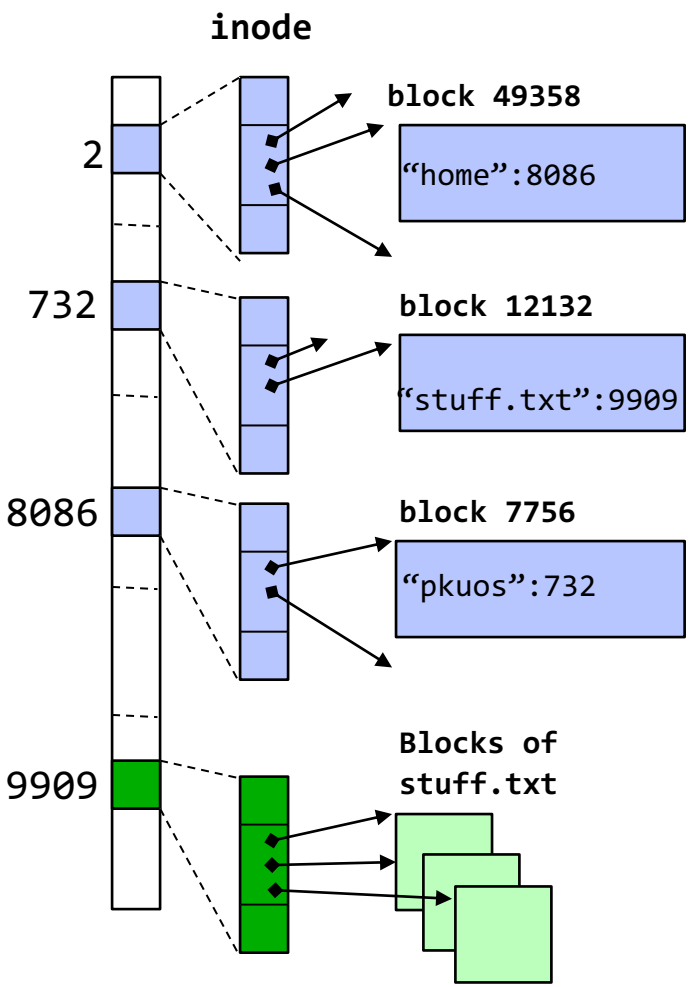
- Contrast these two different types of directory entries:
  - Normal directory entry: <file name, file #>
  - Symbolic link: <file name, dest. file name>

- OS looks up destination file name **each time** program accesses source file name
  - Lookup can fail (error result from **open**)

- Unix: Create soft links with **symlink** syscall

Directory Traversal

- What happens when we open /home/pkuos/stuff.txt?
- "/" - inumber for root inode configured into kernel, say 2
  - Read inode 2 from its position in inode array on disk
  - Extract the direct and indirect block pointers
  - Determine block that holds root directory (say block 49358)
  - Read that block, scan it for "home" to get inumber for this directory (say 8086)
- Read inode 8086 for /home, extract its blocks, read block (say 7756), scan it for "pkuos" to get its inumber (say 732)
- Read inode 732 for /home/pkuos, extract its blocks, read block (say 12132), scan it for "stuff.txt" to get its inumber, say 9909
- Read inode 9909 for /home/pkuos/stuff.txt
- Set up file description to refer to this inode so reads / write can access the data blocks referenced by its direct and indirect pointers
- **Check permissions on the final inode and each directory's inode…**



Large Directories

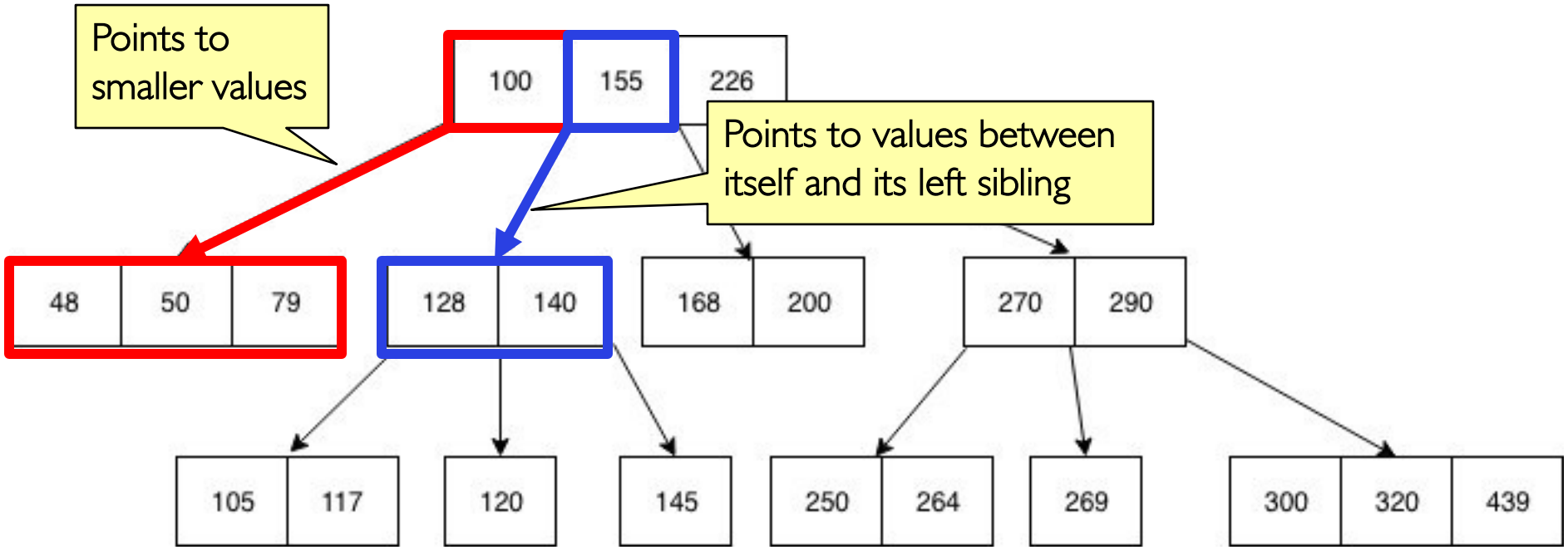
- Early file systems organize directories as:
  - List of <file_name, inode> entries
  - Array of <file_name, inode> entries

- Challenges
  - Linear search: expensive
  - Might need to read entire directory just to find a file: many disk accesses

Large Directories: B-Trees (dirhash)

in FreeBSD, NetBSD, OpenBSD

Recall B-Trees data structure



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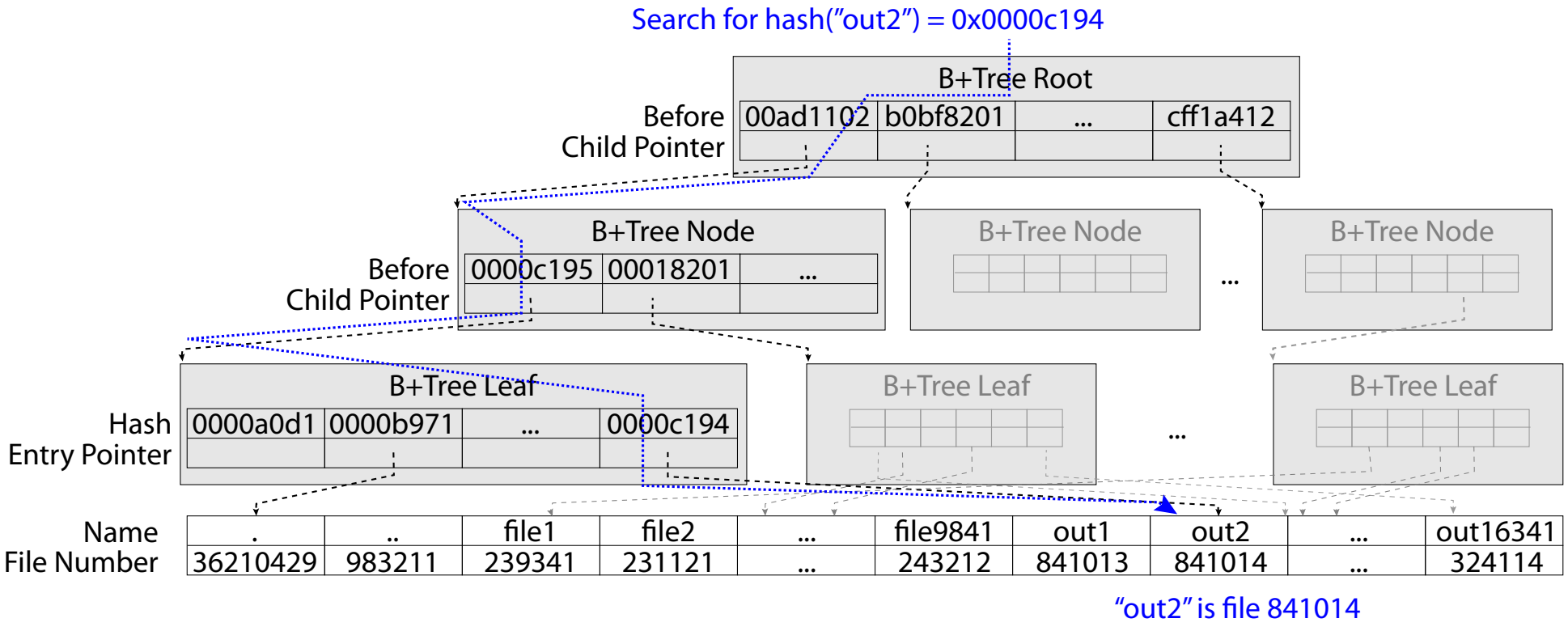
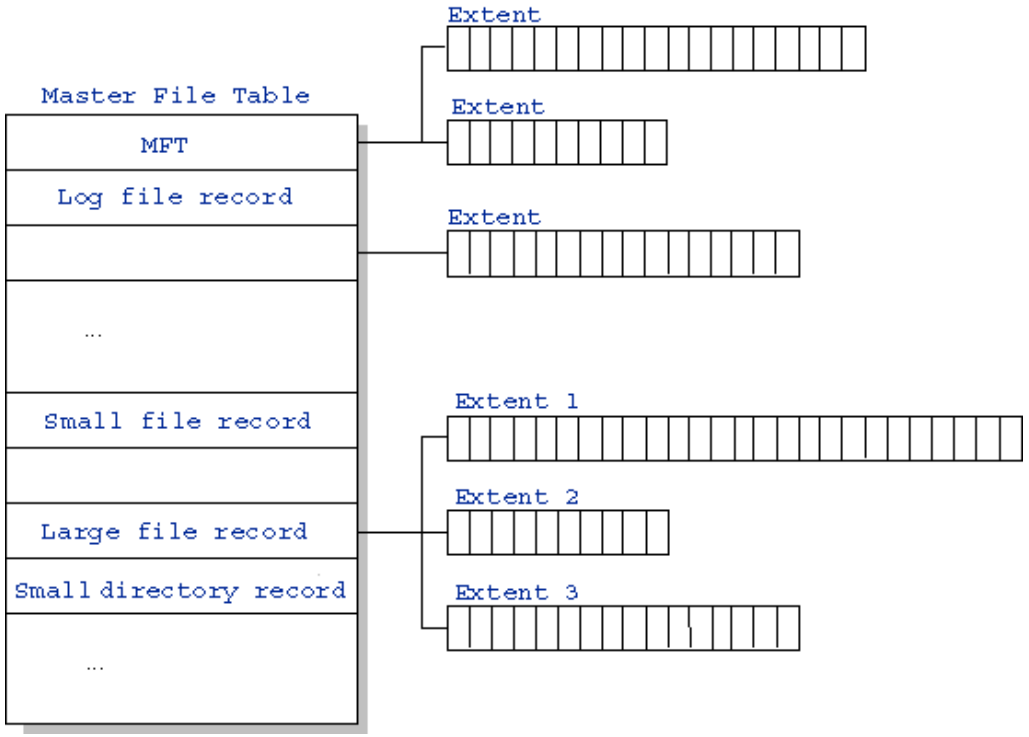


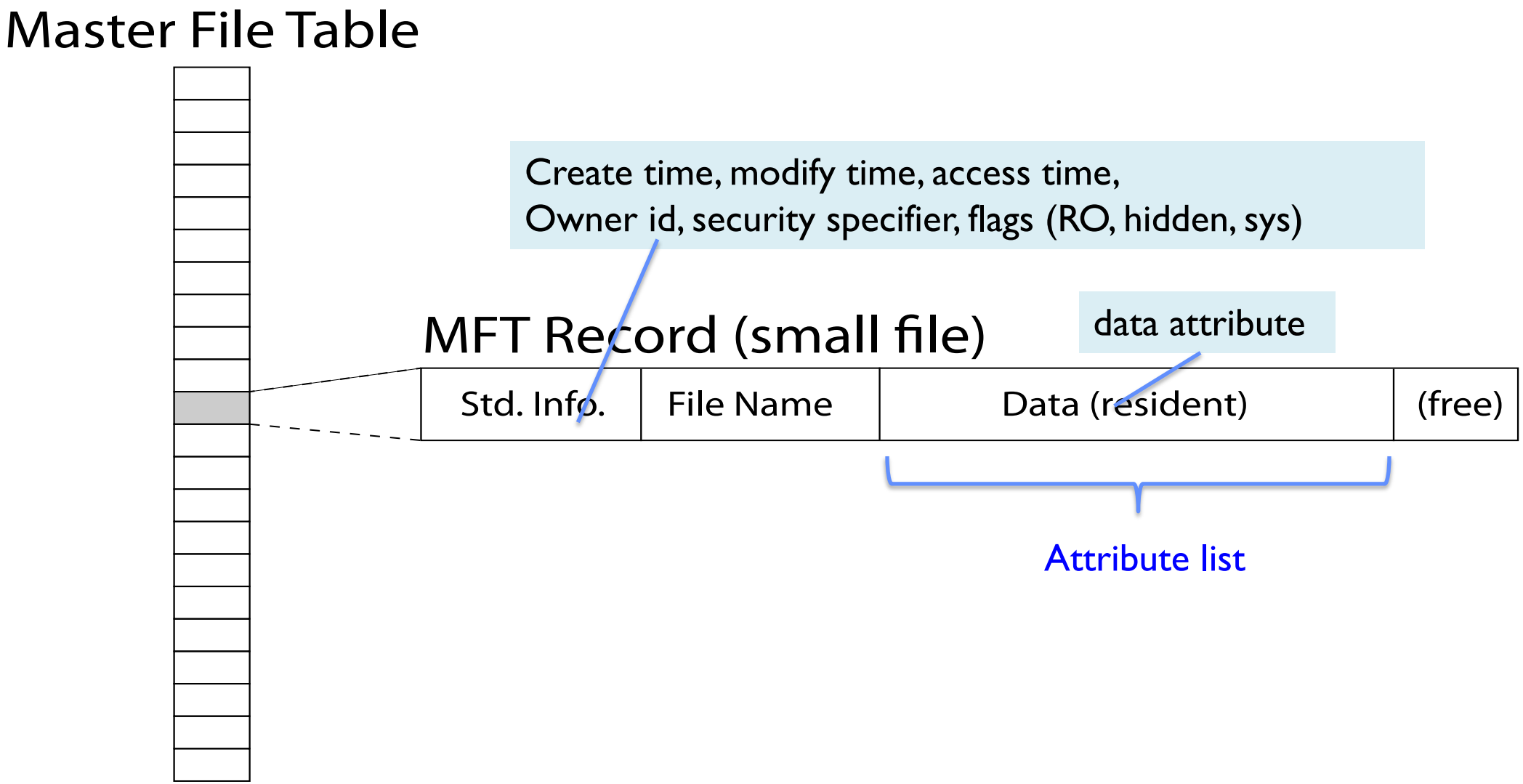
CASE STUDY: WINDOWS NTFS

New Technology File System (NTFS)

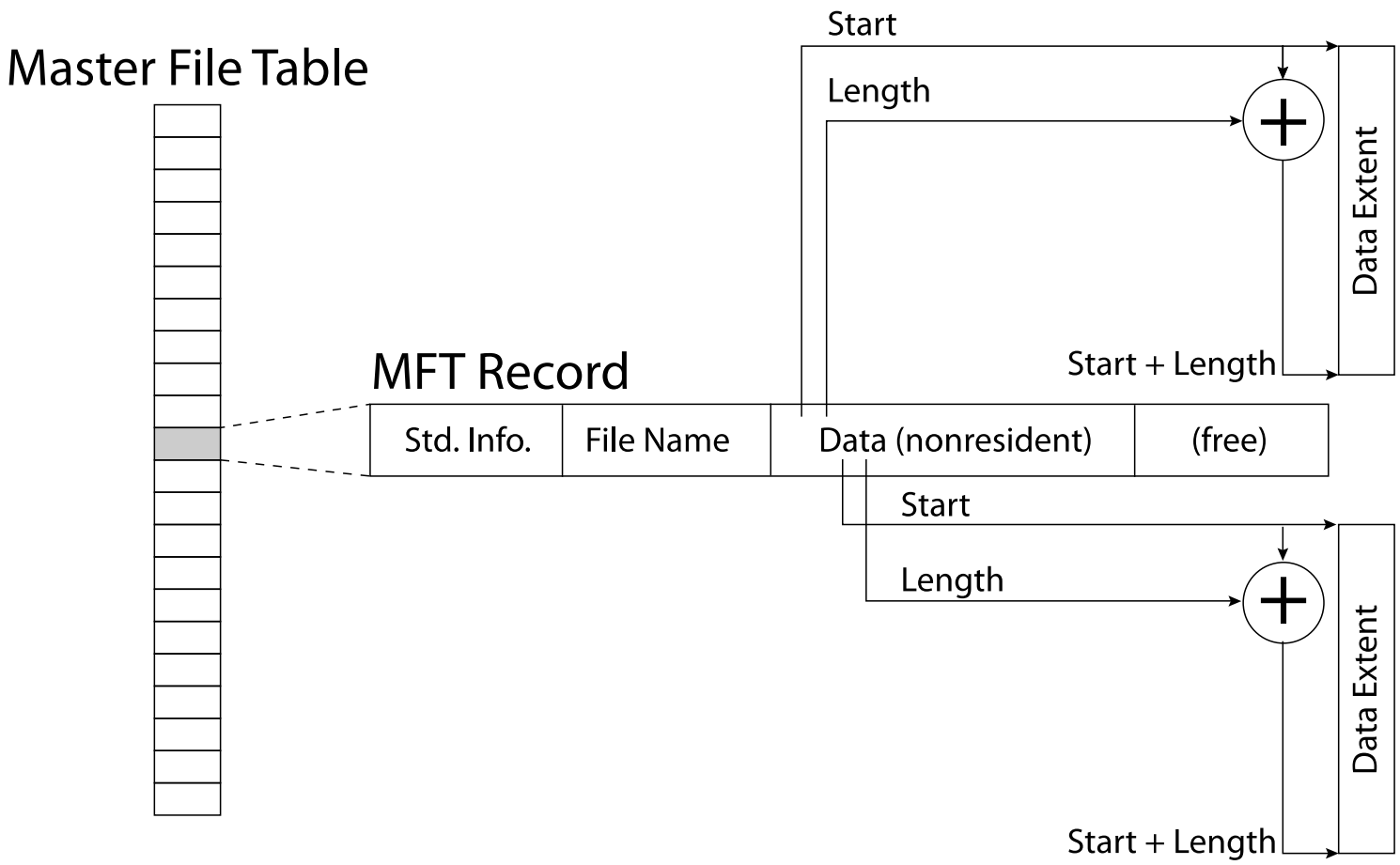
- Default on modern Windows systems
- Variable length extents
  - Rather than fixed blocks
- Instead of FAT or inode array: Master File Table
  - Like a database, with max 1 KB size for each table entry
  - Everything (almost) is a sequence of <attribute:value> pairs
    - » Meta-data and data
- Each entry in MFT contains metadata and:
  - File's data directly (for small files)
  - A list of *extents* (start block, size) for file's data
  - For big files: pointers to other MFT entries with *more* extent lists



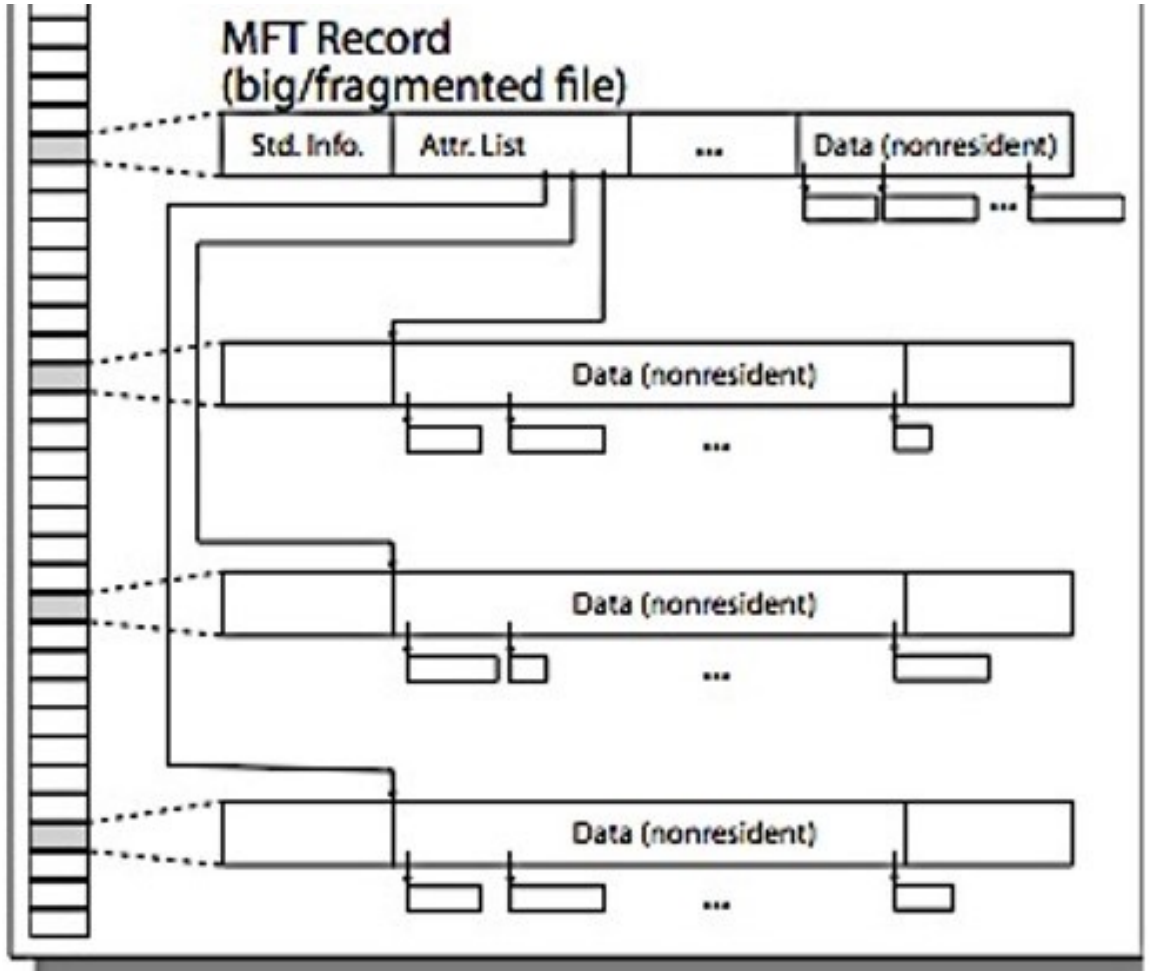
NTFS Small File: Data stored with Metadata



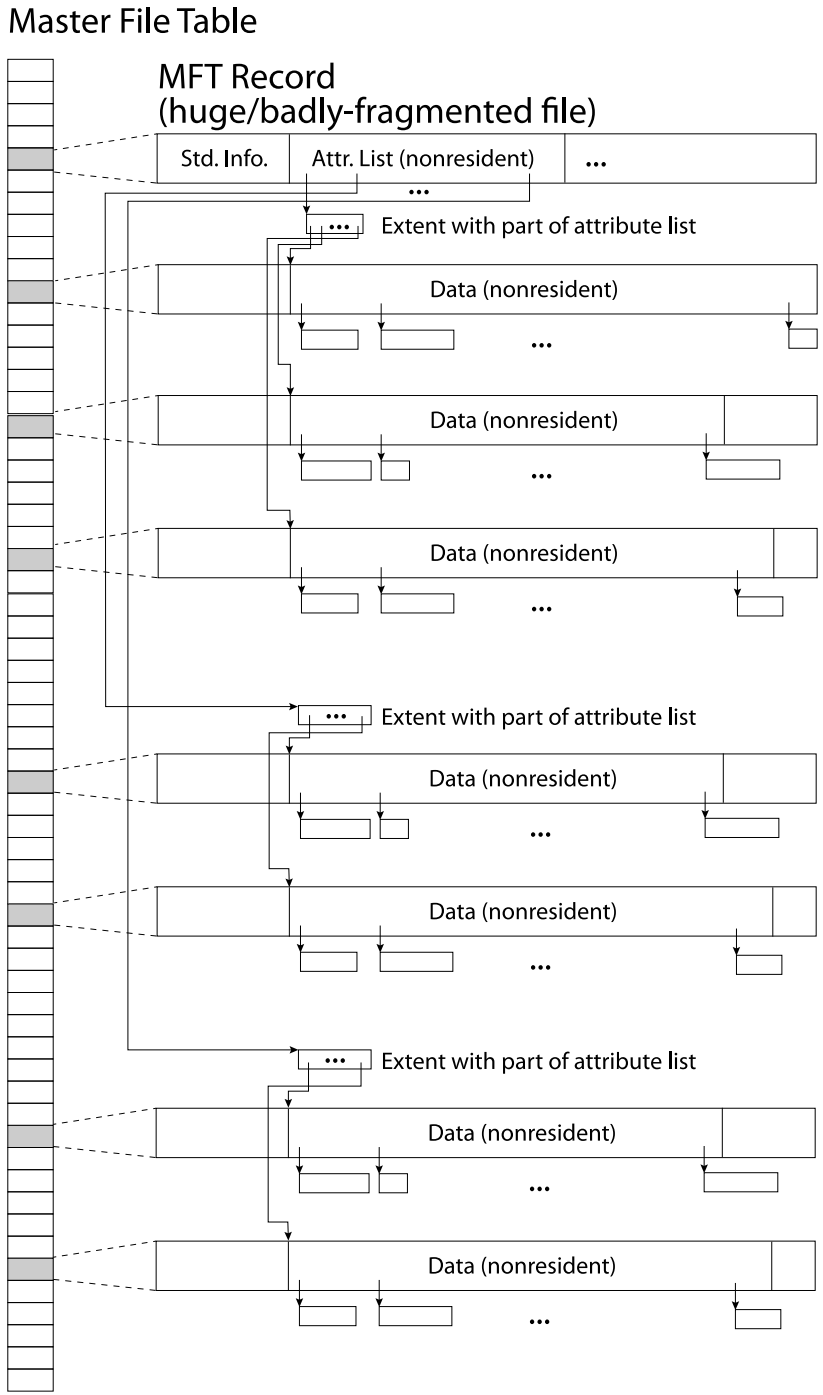
NTFS Medium File: Extents for File Data



NTFS Large File: Pointers to Other MFT Records



NTFS Huge, Fragmented File: Many MFT Records



NTFS Directories

- Directories implemented as B Trees
- File's number identifies its entry in MFT
- MFT entry always has a file name attribute
  - Human readable name, file number of parent dir
- Hard link? Multiple file name attributes in MFT entry

File System Summary (1/2)

- File System:
  - Transforms blocks into Files and Directories
  - Optimize for size, access and usage patterns
  - Maximize sequential access, allow efficient random access
  - Projects the OS protection and security regime (UGO vs ACL)
- File defined by header, called “inode”
- Naming: translating from user-visible names to actual sys resources
  - Directories used for naming for local file systems
  - Linked or tree structure stored in files
- 4.2 BSD Multilevel Indexed Scheme
  - inode contains file info, direct pointers to blocks, indirect blocks, doubly indirect, etc..
  - NTFS: variable extents not fixed blocks, tiny files data is in header

File System Summary (2/2)

- File layout driven by freespace management
  - Optimizations for sequential access: start new files in open ranges of free blocks, rotational optimization
  - Integrate freespace, inode table, file blocks and dirs into block group